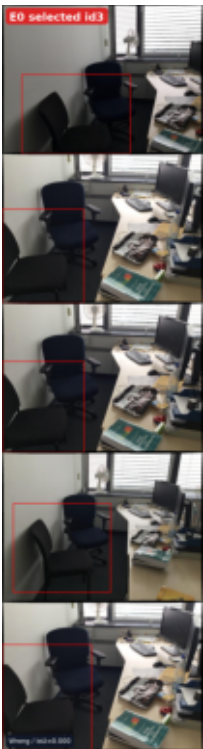
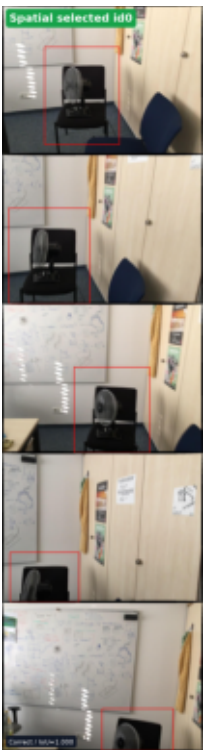


Qualitative 3: E0 Failure vs Spatial-Only Success

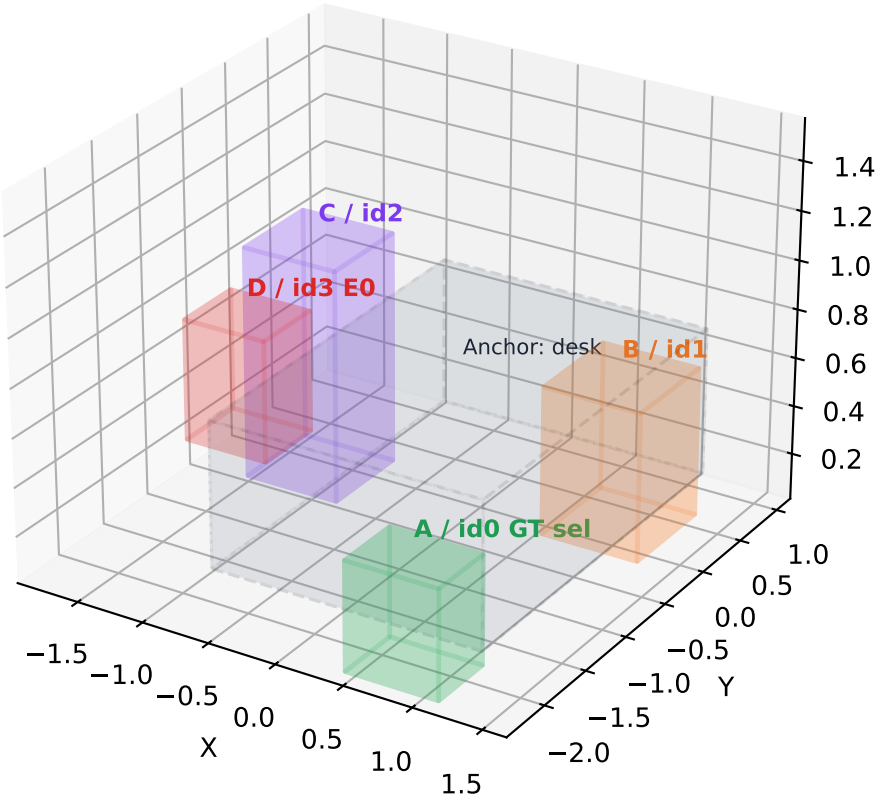
E0 RGB canvas prediction



Ground-truth / spatial-only prediction



Spatial-only evidence



Query
"it is the chair by itself not near the desk"

E0 route
- input: RGB canvases
- selected: id3
- IoU: 0.000 / Acc@0.25=False

Spatial-only route
- input: category + 3D position + anchor distance
- selected: A / id0
- IoU: 1.000 / Acc@0.25=True

Prompt excerpt
Query: "it is the chair by itself not near the desk"
Anchors:
id12 desk: center=(-0.02,-0.20,+0.31) size=(2.03x2.84x0.61)
Candidates:
A. id0 chair: center=(+0.90,-2.27,+0.32) size=(0.67x0.51x0.44)
anchor_delta=(+0.92,-2.07,+0.02) xy_dist=2.26m
B. id1 chair: center=(+1.31,-0.48,+0.57) size=(0.69x0.74x0.59)
anchor_delta=(+1.33,-0.27,+0.26) xy_dist=1.36m
C. id2 chair: center=(-1.45,+0.30,+0.36) size=(0.71x0.71x0.97)
anchor_delta=(-1.43,+0.50,+0.05) xy_dist=1.52m